강의계획서(Syllabus)

[1] 기본 정보(Basic Information)

■강의 정보(Course Information)

개설년도/학기 (Year/Semester)	2026 / 2		개설 캠퍼스 (Campus)	서울(Seoul Campus)	
교과목번호 (Course No.)	50648	분반번호 (Class No.)	01	학점 (Credit)	3
교과목명 (Course Title)	빅 데이터 (INTRODUCTION TO BIG DATA)	강의시간/강의실 (Time/Room)		310관 727호 <강의실>목 7,8,9(310관 727호 <강의실> THU7,8,9)	
이수구분 (Course Classification)	전공(Major)	과목구분 (Lecture Type)		단독강의(Lone-teaching course)	
강의유형 (Course Type)	이론(Theoretical course)	원어강의 여부 (Medium of Instruction)			
공학인증 (ABEEK)	Y	블렌디드유형	-	원격수업유형	-
개설대학 (College)	소프트웨어대학(College of Software)	개설학과(부) (Department)		소프트웨어대학 소프트웨어학부 (School of Computer Science and Engineering)	
e-class 활용여부 (Usage of e-class)	Yes	유연학기			

■RISE 특화교육 연계 강의유형 (RISE Specialized Education: Course Types)

캡스톤디자인 (Capstone Design)	-	프로젝트 기반 (PBL)	-	산학연계교육	-	신산업 융복합	-

■교수자 정보(Instructor Information)

교수명 (Name)	최태영	소속 (Department)	소프트웨어학부(School of Computer Science and Engineering)
연구실전화번호 (Office Phone No.)		연락처 (Contact No.)	0000
E-mail 주소 (E-mail)	tychoe9601@gmail.com	학과전화번호 (Department Phone No.)	5301
상담가능시간 (Office Hour)	목 13:00~15:00	연구실위치 (Office Location)	208-422
홈페이지 (Course Web-site)	-		

[2] 학습 목표/성과(Learning Objectives/Outcomes)

■과목 설명(Course Description)

This course introduces basic concepts and methods in big data analytics, with a special focus on graph-structured data. Students will learn how large-scale data can be collected, represented, analyzed, and interpreted using practical machine learning approaches. The course also introduces Graph Neural Networks as a recent method for analyzing relational data such as social networks, citation networks, and recommendation systems.

■선수과목 및 공통필수과목(Prerequisites and Co-requisites)

■학습 목표(Learning Objectives)

1. Understand the basic concepts of big data and data-driven analysis.
2. Explain common types of graph-structured data and their applications.
3. Understand the basic idea of machine learning for big data analysis.
4. Describe the main concepts behind Graph Neural Networks at an introductory level.

■학습 성과(Learning Outcomes)

1. Identify appropriate data analysis methods for simple big data problems.
2. Represent relational data as graphs and explain basic graph properties.
3. Understand how GNNs can be used for node classification, link prediction, and recommendation tasks.
4. Present analysis results in a clear and understandable way.

[3] 강의 진행 정보(Course Methods)

■강의 진행 방식(Teaching and Learning Methods)

강의 진행 방식(Teaching and Learning Methods)	추가 설명(Additional Description)
강의(Lecture)	

■과제(Assignments)

과제(Assignments)	횟수(No.)	과제 설명(내용, 양식, 분량 등)(Assignments Description)
보고서(Report)	2	A brief report on a selected topic related to graph data analysis and individual project.
개인프로젝트(Individual)	1	An individual project applying the concepts covered in class, including a individual presentation of the results.

■수업 자료(Textbooks, Reading, and other Materials)

수업 자료 (Textbook/Reference)	제목(Title)	저자(Author)	출판일/게재일 (Year of Publication/etc)	출판사/학회지 (Publisher/Name of Journal)	판차사항/권호정보(No. of Edition)
주교재(Main Textbook)	Graph Representation Learning	William L. Hamilton	2020	Morgan & Claypool Publishers	

[4] 학습 평가 방법(Student Assessment)

평가 항목(Assessment Item)	평가 비율 (%)(Assessment Ratio)	추가 설명(Additional Description)
출결(Attendance)	5	
중간시험(Mid-term Exam)	35	

평가 항목(Assessment Item)	평가 비율 (%)(Assessment Ratio)	추가 설명(Additional Description)
과제(Assignment)	60	

[5] 수업 일정(Course Schedule)

주(Week)	강사명 (Instructor)	수업주제 및 내용(Topic & Content)	학습과제 (Student Assignment)	추가설명 및 교수과제 (Additional Description & Instructor Assignment)
1	최태영	Introduction		
2	최태영	Fundamental of Big data (1)		
3	최태영	Graph Representation Basics		
4	최태영	Link Analysis and PageRank		
5	최태영	Understanding of Graph Embedding		
6	최태영	Graph Neural Networks (1)	Report 1	Proposal of individual project.
7	최태영	Graph Neural Networks (2)		
8	최태영	Mid-term exam		
9	최태영	Graph Neural Networks (3)		
10	최태영	Self-Supervised and Contrastive Learning on Graphs		
11	최태영	GNN for Recommendation	Report 2	
12	최태영	Community and Subgraph Analysis		
13	최태영	Generative Models for Graphs		
14	최태영	Individual presentation		Final presentation of individual project.
15	최태영	Individual presentation (2)		
16	최태영	Final exam		

[6] 수강생 학습 안내 사항(Guide to Learning)

Problem solving with data centric approaches.

기출문제(Previous Exam Samples)

<추가 자료 다운로드>(<Download Additional Sample>)

첨부 자료가 없습니다.

공학 교육 인증용 설계교과목(Engineering Education)

프로그램 학습 성과(Learning Outcomes)

지식응용 : 30 **문제해결** : 10 **도구활용** : 30 **설계능력** : 30

■설계 과목 제목(Title)

■설계 과제 목표(Objective)

■설계 제한 사항(Restrictions)

■설계 평가 방법(Assessment Method)

※시험 부정 행위 적발 시 중앙대학교 학칙 제71조 【징계】및 학사운영규정 【제 6장 제 47조 【시험 부정행위자의 처벌 】에 따라 징계 처분 대상이 될 수 있습니다.
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1. 시각장애 : 점자, 확대자료, 파일자료, 대필도우미 배치, 평가방법 조정 등
2. 청각장애 : 대필도우미, 속기사 배치, 평가방법 조정 등
3. 지체/뇌병변장애 : 휠체어접근이 가능한 강의실 제공, 대필도우미 배치, 평가방법 조정 등
4. 기타 정신적 장애, 건강장애의 경우 장애학생지원센터 상담 후 지원
※문의처 : 02-820-6577~9(서울캠퍼스), 031-670-4816(안성캠퍼스)
장애학생지원센터 카카오톡 플러스친구(중앙대학교 장애학생지원센터 또는 cauable)

In this class, students with disabilities are eligible for reasonable accommodations depending on the type and severity of disability. If you wish to receive accommodations listed below, please contact the Support Center for Students with Disabilities.
1. Visual Impairment: Braille, large print, electronic class materials, volunteer note-taker, adjustments in assessment practices, etc.
2. Hearing Impairment: Volunteer note-taker, stenographer, adjustments in assessment practices, etc.
3. Physical Disabilities/Brain Lesions: Classrooms with wheelchair access, volunteer note-taker, adjustments in assessment practices, etc.
4. Accommodations for students with other psychiatric disabilities or health impairments can be arranged through the Support Center for Students with Disabilities after consultation.
※Inquiry: 02-820-6577~9 (Seoul Campus), 031-670-4816 (Anseong Campus)
- KakaoTalk Plus Friend ID: @cauable